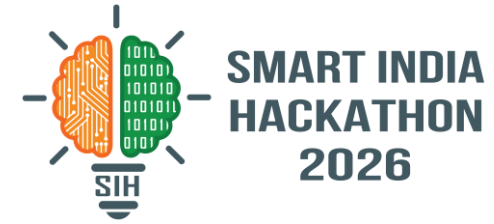
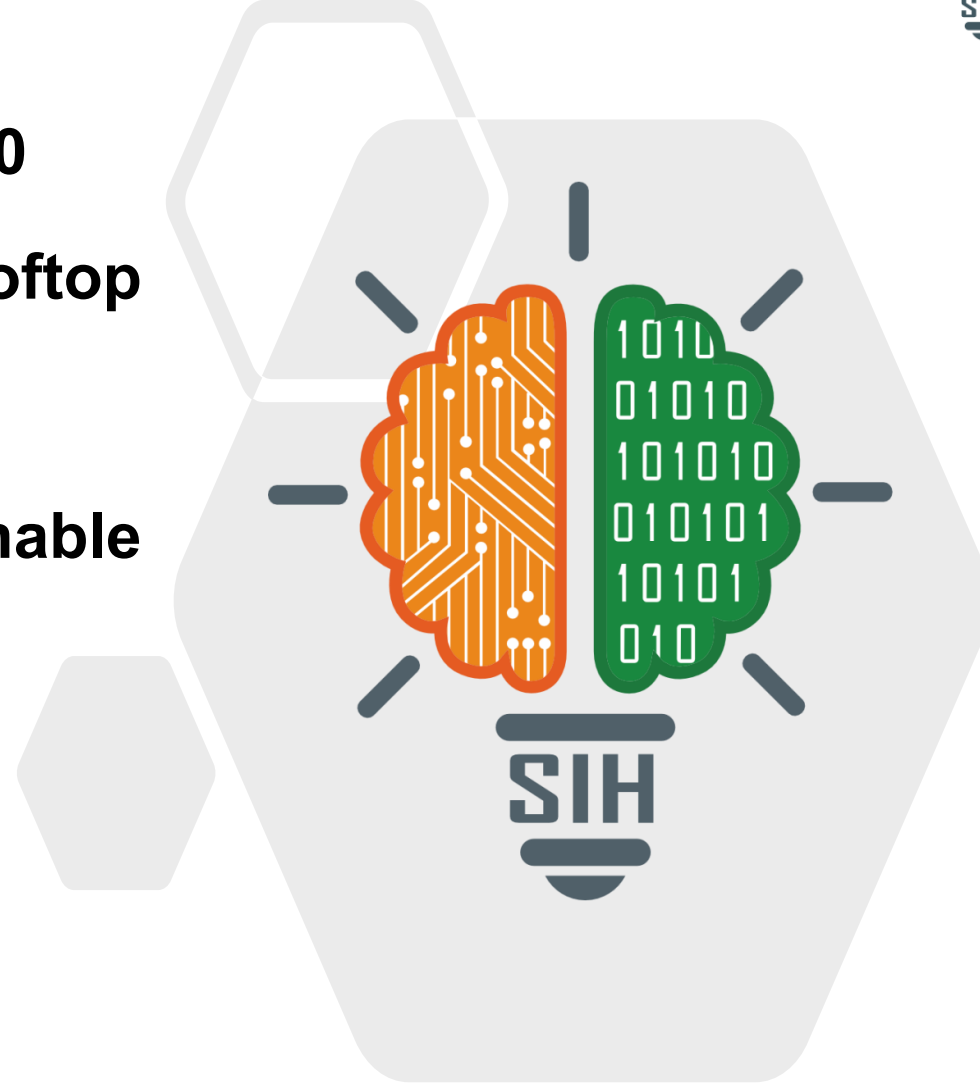


SMART INDIA HACKATHON 2026



- **Problem Statement ID – SIH26200**
- **Problem Statement Title- Rooftop Solar Estimator.**
- **Theme- Renewable/Sustainable Energy.**
- **PS Category- Software**
- **Team ID- SIH262**
- **Team Name- INNOVEXA**

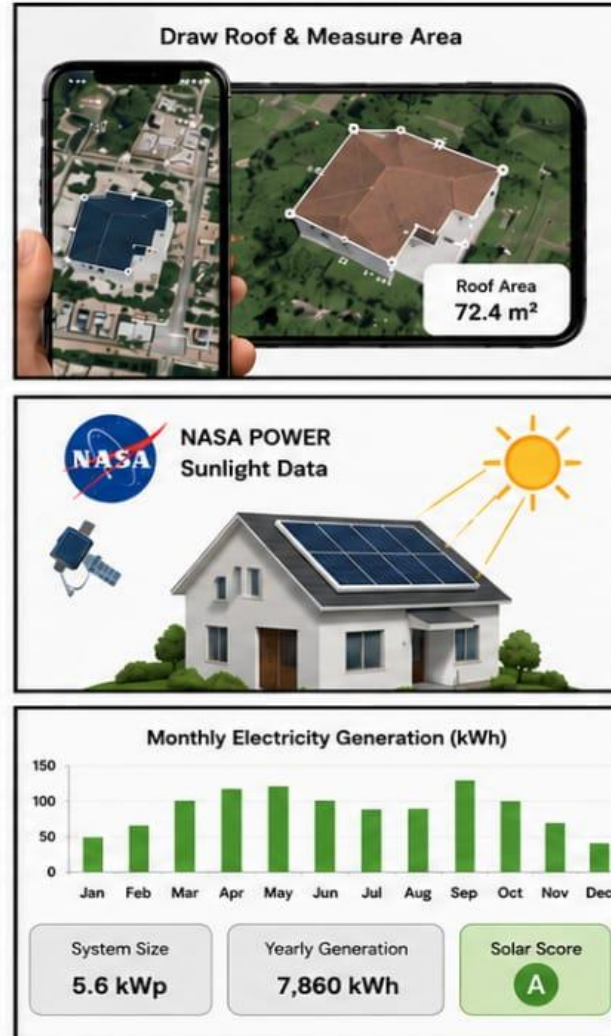


SOLAR SAATHI

❖ Proposed Solution



METHODOLOGY & PROCESS OF IMPLEMENTATION



TECHNOLOGIES TO BE USED



Web app (HTML, CSS, JavaScript)
Runs in any smartphone browser, no installation



Leaflet Maps
Satellite imagery to find and draw the roof



NASA POWER API
Free satellite sunlight data for the exact location (no API key needed)



JavaScript Calculation Engine
Panels, generation, savings, subsidy and payback



Hosted Free
Netlify / Vercel / Replit



Hardware
Only a smartphone – no special hardware, no site visit



ANALYSIS OF FEASIBILITY



Built using basic web technologies



Users manually outline rooftops using an interactive satellite map



Location-specific solar data can be obtained from NASA POWER



PM Surya Ghar subsidy and loan calculations can be integrated into the system



Panel presets allow comparison of different solar technologies



No specialised hardware is required



POTENTIAL CHALLENGES & RISKS



Satellite images may not clearly show every rooftop



Roof shading and orientation can affect actual generation



Live map or solar data may occasionally be unavailable



STRATEGIES FOR OVERCOMING CHALLENGES



Manual roof drawing gives users control over boundary selection



Results clearly shown as estimates with editable assumptions



Offline fallback estimation available when live data fails



Demo Roof mode ensures uninterrupted prototype demonstration



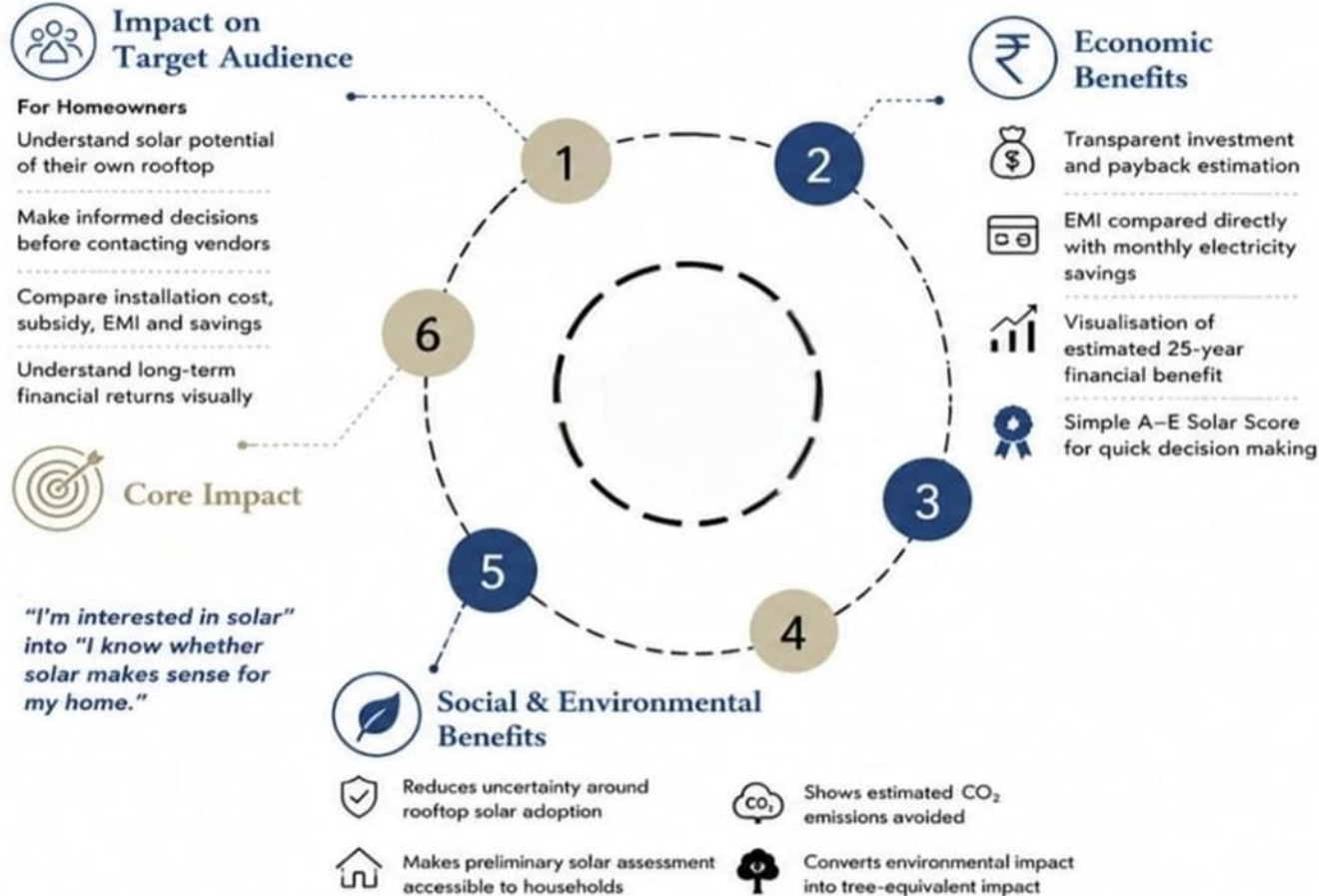
Future scope: AI-based roof detection and shading analysis



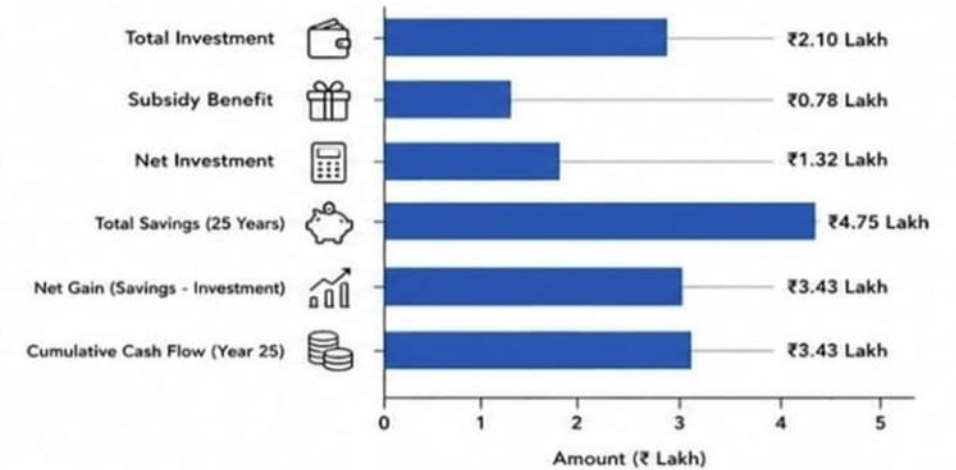
BOTTOM HIGHLIGHT

Technically feasible as a low-cost prototype using open data and scalable for future enhancements.



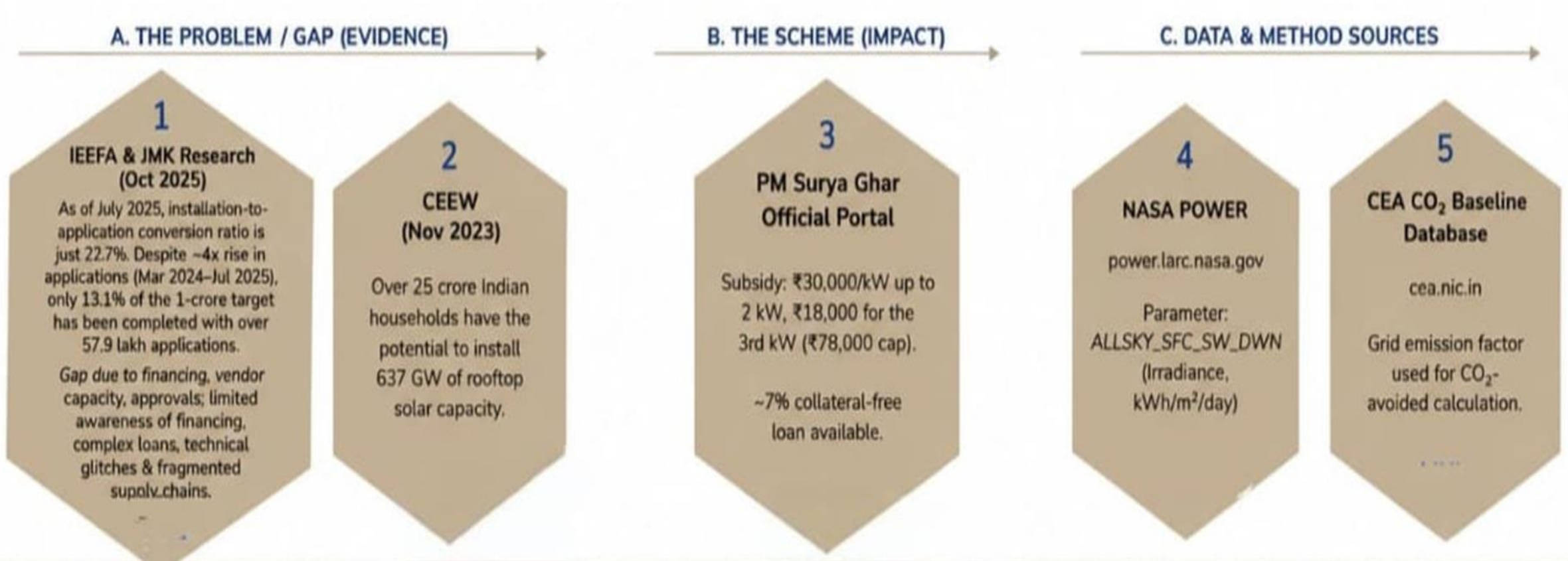


25-Year Financial Benefit Visualization



From Grids to Green Impact





1 :- <http://ieefa.org/resources/advancing-residential-rooftop-solar-adoption-india-under-pm-surya-ghar-yojana>

3 :- <http://pmsuryaghar.gov.in/>

4 :- <http://power.larc.nasa.gov/>

5 :- <http://cea.nic.in/>